FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

AI-POWERED STOCK SCREENING AND ANALYSIS

A PROJECT REPORT

Submitted by

240701092 – DANYL ENOCH

240701329 – MOHIDEEN ABDUL KADER M L

Academic Year 2026–2027



BONAFIDE CERTIFICATE

Certified that this project report “AI-Powered Stock Screening and Analysis” is the bonafide work of “Danyl Enoch” and “Mohideen Abdul Kader M L” who carried out the project work under my supervision.

SIGNATURE OF THE FACULTY INCHARGE	
Submitted for the Practical Examination held on _____	
SIGNATURE OF THE INTERNAL EXAMINER	

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	4
1	INTRODUCTION	5
2	LITERATURE REVIEW	7
3	PROPOSED SYSTEM	9
3.1	ARCHITECTURE DIAGRAM	10
4	MODULES DESCRIPTION	12
4.1	FUNDAMENTAL ANALYSIS MODULE	12
4.2	TECHNICAL ANALYSIS MODULE	13
4.3	SENTIMENT ANALYSIS MODULE	14
4.4	AI / ML ENGINE MODULE	15
5	IMPLEMENTATION AND RESULTS	17
5.1	EXPERIMENTAL SETUP	17
5.2	RESULTS	19
6	CONCLUSION AND FUTURE WORK	21
7	REFERENCES	22

ABSTRACT

Investors today must evaluate large volumes of stocks before making sound investment decisions, but traditional analysis demands in-depth manual study of financial statements, historical prices, market trends, and technical indicators. This manual evaluation, spread across fragmented tools and data sources, is timeconsuming and complex, particularly for beginners.

This project proposes an intelligent, unified system that leverages artificial intelligence and data-driven techniques to screen and analyze equities efficiently. The system integrates fundamental analysis, technical analysis, and news sentiment analysis using Natural Language Processing (NLP) into a single interface. It computes automated scores, rankings, and plain-language explanations for each stock.

At the core of the analytical engine is the Steepest-Ascent Hill Climbing algorithm, a local-search optimization technique that iteratively moves toward the neighboring state offering the steepest improvement in score. In this project, the algorithm is used to optimize stock-scoring weights toward strong-performing stock rankings.

The proposed system is intended as a scalable and explainable decision-support tool that reduces manual screening effort and makes multi-factor analysis more accessible to investors of different experience levels. The system provides analytical insight and is not intended to guarantee returns or function as financial advice.

1 INTRODUCTION

Stock screening is the process of evaluating a collection of stocks using selected financial, market, and other criteria in order to identify candidates for further analysis. Traditional screening often requires the investor to inspect financial statements, historical prices, technical indicators, market trends, and financial news separately.

The project addresses this fragmented workflow by proposing a unified AIpowered stock screening and analysis system. Instead of requiring users to move between several single-purpose tools, the proposed system brings fundamental, technical, and sentiment analysis together in one interface.

1.1 Need for the System

• Large volumes of stock information make manual evaluation time-consuming.

- Financial information is distributed across different types of sources and tools.

• Beginners may find multi-factor analysis complex to perform manually. • A unified interface can combine multiple analytical perspectives. • Automated scoring and plain-language explanations can make results easier to compare.

1.2 Aim

The aim is to develop an intelligent, unified decision-support system for stock screening and analysis by combining fundamental, technical, and sentiment analysis and optimizing the scoring process with Steepest-Ascent Hill Climbing.

1.3 OBJECTIVES

1. Provide a unified interface for stock screening and analysis. 2. Collect market prices, financial statements, trading volume, and news text for analysis.

3. Process and normalize collected information before analysis. 4. Perform fundamental analysis using financial ratios, earnings, and balancesheet metrics.
5. Perform technical analysis using Moving Averages, RSI, MACD, and price/volume dynamics.
6. Perform sentiment analysis using NLP and classify text as positive, neutral, or negative.
7. Combine analytical features through an AI/ML engine.
8. Use Steepest-Ascent Hill Climbing to optimize stock-scoring weights.
9. Generate stock scores, rankings, charts, indicators, and AI-generated explanations.

1.4 Scope

The project focuses on analytical and decision support. Its scope covers screening and multi-factor analysis of equities, integration of three analytical branches, optimization of scoring weights, and presentation of results through a responsive dashboard. The project content specifically describes the system as providing insight rather than guaranteed returns or financial advice.

1.5 Organization of the Report

The report presents the background and literature review, proposed system and architecture, individual modules, implementation and outputs, followed by conclusion, future work, and references.

2 LITERATURE REVIEW

The project literature survey identifies several established approaches to stock screening and analysis. These approaches form the conceptual basis for combining multiple analytical branches within one decision-support system.

Area	Approach described in the project
Stock Screening Systems	Rule-based filtering using Market Capitalization, P/E, P/B, revenue growth, and dividend yield.
Technical Analysis	Price action and volume trends using Moving Averages, RSI, and MACD.
Machine Learning in Stock Analysis	Predictive and classification models such as Linear Regression, Decision Trees, and Random Forest.
Sentiment Analysis	NLP analysis of financial news, corporate reports, and media text.

Steepest-Ascent Hill Climbing	Local-search optimization that moves to the neighboring state with the steepest score improvement.

2.1 Stock Screening Systems

Traditional screeners use rule-based metrics to narrow a larger set of assets into a smaller candidate set. The project identifies Market Capitalization, P/E, P/B, revenue growth, and dividend yield as examples of such metrics.

2.2 Technical Analysis

Technical analysis examines historical price and volume behavior. The project

specifically identifies Moving Averages, Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD) as technical indicators used to evaluate historical patterns.

2.3 MACHINE LEARNING IN STOCK ANALYSIS

The literature survey in the project identifies predictive and classification models as approaches for identifying underlying non-linear patterns in financial data. Examples listed are Linear Regression, Decision Trees, and Random Forest.

2.4 SENTIMENT ANALYSIS USING NLP

Financial news, corporate reports, and media text can provide information that is different from numerical market indicators. The project uses Natural Language Processing to classify textual sentiment into positive, neutral, or negative categories.

2.5 STEEPEST-ASCENT HILL CLIMBING

Steepest-Ascent Hill Climbing is presented as a local-search optimization technique. At each stage, the method considers neighboring states and moves toward the state that provides the steepest improvement in the score. In the proposed stock analysis system, this approach is used to optimize stock-scoring weights.

2.6 RESEARCH GAP

The project identifies a gap in existing workflows: users may rely on manual filters across disconnected tools, while explainable AI is not integrated into the same workflow. The proposed solution addresses this stated gap by combining screening, technical analysis, fundamental analysis, and AI-driven explanations within one platform.

3 PROPOSED SYSTEM

The proposed system is an AI-powered stock screening and analysis platform designed as a unified decision-support environment. The system receives stock tickers, screening preferences, and custom filter criteria from the user.

3.1 Main Processing Flow

The system follows a seven-layer pipeline. User inputs are received through the user layer and interface. Relevant market and textual information is collected, processed, and passed into three analysis branches. The resulting features are combined by the AI/ML engine, after which stock scores, rankings, charts, indicators, and explanations are presented.

Stage	**Main responsibility**
Input	Stock tickers, screening preferences, custom filters
Interface	Responsive web/mobile dashboard
Collection	Market prices, financial statements, volume, news text
Processing	Cleaning, normalization, feature engineering
Analysis	Fundamental, technical, sentiment branches
AI/ML	Feature combination, pattern analysis, scoring/ranking
Output	Scores, charts, indicators, AI explanations

3.2 Design Characteristics

• Unified analysis rather than disconnected single-purpose tools.
• Multi-branch analysis using fundamental, technical, and sentiment information. •

Optimization of stock-scoring weights using a local-search technique. • Explainable output through plain-language summaries.

• Responsive presentation through a web/mobile dashboard.

3.1 ARCHITECTURE DIAGRAM

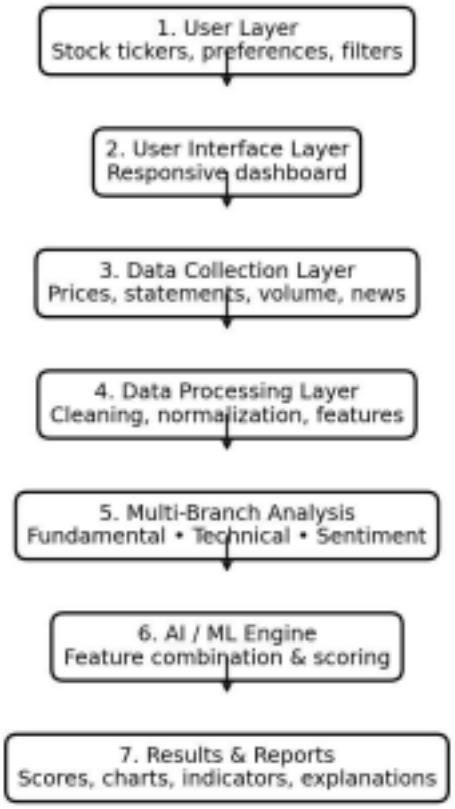


Figure 3.1 Seven-layer architecture of the proposed AI-powered stock screening and analysis system.

3.1.1 Layer Description

The User Layer accepts stock tickers, screening preferences, and custom filter criteria. The User Interface Layer provides the dashboard through which users interact with the system and visualize results.

The Data Collection Layer aggregates market prices, financial statements, trading volume, and news text. The Data Processing Layer performs data cleaning, normalization, and feature engineering.The Multi-Branch Analysis layer runs fundamental, technical, and sentiment analysis in parallel. The AI/ML Engine combines features, performs pattern analysis, and generates stock scores and

rankings. Finally, the Results & Reports Layer presents scores, comparative charts, indicators, and AI-generated explanations.

3.2 DATA COLLECTION AND PROCESSING

3.2.1 Data Collection Layer

The project architecture specifies collection of market prices, financial statements, trading volume, and news text. These inputs provide the numerical and textual information needed by the three analysis branches.

3.2.2 Data Processing Layer

Before analysis, collected information passes through data cleaning, normalization, and feature engineering. The purpose of this layer in the architecture is to prepare the collected information so that it can be combined by the analytical engine.

3.2.3 Multi-Branch Analysis

After processing, three branches operate in parallel. Fundamental analysis examines company-level financial information. Technical analysis evaluates historical price and volume behavior. Sentiment analysis processes financial text using NLP.

3.2.4 Feature Combination

The outputs of the three branches are combined by the AI/ML engine. This provides a multi-factor representation rather than depending on a single analytical signal.

4 MODULES DESCRIPTION

4.1 FUNDAMENTAL ANALYSIS MODULE

The Fundamental Analysis Module evaluates financial ratios, earnings, and balance-sheet metrics to gauge a company's underlying financial health.

Component	Role described in project
Financial ratios	Provide quantitative measures for evaluating the company.
Earnings	Represent an important part of company-level financial evaluation.
Balance-sheet metrics	Provide information for assessing underlying financial health.
Fundamental features	Become inputs to the combined AI/ML scoring process.

4.1.1 Processing Role

The module belongs to the first analytical branch of the system. Its outputs are processed features rather than a standalone final decision. They are later combined with technical and sentiment features.

4.1.2 Contribution to Stock Scoring

Fundamental features contribute to the composite weighted score generated by the system. The project architecture indicates that the AI/ML engine combines the outputs from all three branches before generating stock scores and rankings.

4.2 TECHNICAL ANALYSIS MODULE

The Technical Analysis Module tracks moving averages, RSI, MACD, and price/volume dynamics to evaluate historical trading patterns.

Indicator / Input	Purpose in the project

Moving Averages	Track historical price trends.
RSI	Used as a technical indicator for evaluating historical patterns.

MACD	Used as a technical indicator for evaluating historical patterns.
Price and volume dynamics	Provide historical market behavior for analysis.

4.2.1 Processing

Historical market information enters the data processing layer, where it is cleaned, normalized, and transformed into features. The technical analysis branch then evaluates the specified indicators and price/volume dynamics.

4.2.2 Output

Technical features become part of the combined analytical representation used by the AI/ML engine. The final dashboard can surface key technical indicators alongside stock scores and other information.

4.3 SENTIMENT ANALYSIS MODULE

The Sentiment Analysis Module applies Natural Language Processing (NLP) to financial news, corporate reports, and media text. Its purpose is to classify textual information as positive, neutral, or negative.

4.3.1 Input

• Financial news

• Corporate reports Media text

4.3.2 NLP-Based Classification

The module converts textual information into sentiment information that can be

incorporated into the broader stock analysis process. The project specifically defines three sentiment classes: positive, neutral, and negative.

4.3.3 Contribution to the System

Sentiment analysis adds a textual perspective to the numerical information provided by fundamental and technical branches. The resulting sentiment features are passed to the AI/ML engine for feature combination and stock scoring.

4.3.4 Explainability

The final system is designed to provide plain-language explanations for stock scores. Sentiment information is one of the analytical inputs contributing to this multi-factor explanation.

4.4 AI / ML ENGINE MODULE

The AI/ML Engine Module combines features from the fundamental, technical, and sentiment branches. The project describes the engine as using Random Forest, RNN/sentiment models, classification, and pattern analysis, with optimization through Steepest-Ascent Hill Climbing.

Component	**Role**
Feature combination	Combines outputs from the three analysis branches.
Random Forest	Listed as an analytical/modeling approach in the engine.
RNN / sentiment models	Listed for sentiment-related modeling.
Classification	Supports analytical categorization.
Pattern analysis	Identifies patterns in the combined feature space.
Hill Climbing	Optimizes stock-scoring weights.

4.4.1 Stock Scoring

The engine produces a composite, weighted score for each stock. The scores are then used to generate rankings that make candidate comparison easier.

4.4.2 Explainable Output

In addition to numerical scores and rankings, the system generates plain-language summaries explaining why a stock received its score. This supports the project's stated goal of explainable decision support.

4.5 STEEPEST-ASCENT HILL CLIMBING

Steepest-Ascent Hill Climbing is a local-search optimization technique. The project uses it to optimize the weights used in stock scoring.

4.5.1 Basic Principle

A state represents a particular configuration of scoring weights. Neighboring states represent alternative weight configurations. The algorithm evaluates the available neighboring states and moves to the neighbor that provides the steepest improvement in the score.

4.5.2 Conceptual Steps

10. Start with an initial scoring-weight configuration.
11. Generate or consider neighboring weight configurations.
12. Evaluate the score associated with each neighboring state.
13. Select the neighbor with the steepest improvement.
14. Move to that state.
15. Repeat until no neighboring state provides an

improvement.

4.5.3 Role in the Proposed System

Within the proposed architecture, hill climbing is positioned in the AI/ML engine. Its purpose is not to collect data or directly perform the individual fundamental, technical, or sentiment analyses; instead, it optimizes the scoring weights used to combine analytical features.

5 IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The project presentation describes the system as a responsive web/mobile dashboard backed by a multi-layer analytical pipeline. The experimental setup therefore follows the architecture presented in the project material.

Setup Element	**Description from project**
User interface	Responsive web / mobile dashboard

Inputs	Stock tickers, preferences, custom filter criteria
Data	Market prices, financial statements, trading volume, news text
Processing	Cleaning, normalization, feature engineering
Analysis	Fundamental, technical, sentiment
AI/ML	Feature combination, scoring/ranking, optimization
Output	Scores, rankings, charts, indicators, explanations

5.1.1 End-to-End Operation

A user supplies stock tickers and screening preferences. The system collects relevant numerical and textual information, processes it, runs the three analysis branches, and combines their features. The AI/ML engine generates a composite

score and ranking, while the final dashboard presents the result in an accessible form.

5.1.2 ANALYSIS FLOW

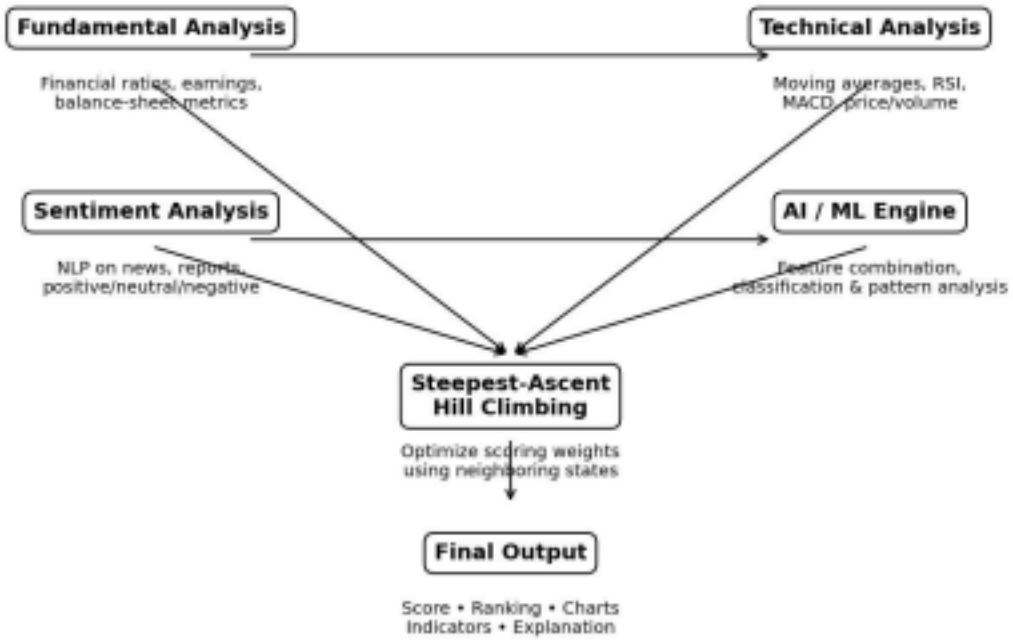


Figure 5.1 Conceptual flow of the analysis modules and final output.

5.1.3 Integrated Decision-Support Flow

The integrated flow begins with the three analytical branches. Fundamental analysis contributes financial-health information, technical analysis contributes historical market-pattern information, and sentiment analysis contributes classified textual information. These branches feed the AI/ML engine, where features are combined and scoring weights are optimized.

The final stage presents the combined result through stock scores, rankings, charts, indicators, and AI-generated explanations.

5.2 RESULTS

5.2.1 Stock Score and Ranking

The system produces a composite, weighted score for each stock and ranks stocks for easy comparison. The project identifies this as one of the principal outputs of

the system.

5.2.2 Dashboard Charts

The dashboard provides visual comparisons of price trends, technical indicators, and multi-factor scores. These visual outputs help users inspect the information generated by the analytical modules.

5.2.3 Indicators

Key technical and fundamental indicators are surfaced alongside each result. This keeps important analytical information available together with the generated score.

5.2.4 AI-Generated Explanations

The system generates plain-language summaries explaining why a stock received its score. This is a central part of the project's explainable decision-support objective.

Output	Description
Stock Score / Ranking	Composite weighted score and comparative ranking.
Dashboard Charts	Visual comparison of trends, indicators, and multi-factor scores.
Indicators	Key technical and fundamental indicators.
AI Explanations	Plain-language reasoning associated with the score.
User Dashboard	Unified visualization of the complete result.

5.3 ADVANTAGES, LIMITATIONS AND APPLICATIONS

5.3.1 Advantages

• Unifies multiple analysis types in a single platform.

• Reduces the need for manual screening across disconnected tools. • Combines numerical and textual information.

• Uses optimization for stock-scoring weights.

• Provides rankings, visualizations, indicators, and explanations. • Designed to be accessible through a responsive dashboard.

5.3.2 Limitations

• The project is a decision-support system and does not guarantee investment returns.

• The quality of outputs depends on the quality and completeness of collected data.

• Hill Climbing is a local-search technique and therefore operates through neighboring-state improvement rather than an exhaustive global search. • The project material describes analytical functionality but does not provide measured accuracy or financial-return benchmarks.

5.3.3 Applications

• Stock screening and candidate comparison.

• Multi-factor equity analysis.

• Educational exploration of AI in financial technology.

• Decision support for users who want consolidated analytical information.

6 CONCLUSION AND FUTURE WORK

6.1 Conclusion

The AI-Powered Stock Screening and Analysis project proposes a unified decision-support system that combines fundamental analysis, technical analysis, and sentiment analysis. Its seven-layer architecture connects user input, interface, data collection, processing, multi-branch analysis, AI/ML processing, and final reporting.

The project uses Steepest-Ascent Hill Climbing to optimize stock-scoring weights. By combining analytical features and producing composite scores, rankings, dashboard charts, indicators, and AI-generated explanations, the proposed system aims to reduce manual screening effort and make multi-factor stock analysis more accessible.

The project is positioned as analytical and decision support rather than financial advice. Its purpose is to provide structured insight and explainable analysis rather than guarantee returns.

6.2 Future Work

The provided project material does not specify a detailed future-work implementation plan. Based strictly on the stated architecture and objectives, future development can focus on extending the unified analytical workflow, improving the data-processing pipeline, enriching explainability, and evaluating the scoring approach using clearly defined experimental measures. These are development directions rather than results already established by the project.

6.3 Final Statement

The proposed system demonstrates how artificial intelligence concepts, NLP, technical and fundamental indicators, and local-search optimization can be combined into a single financial technology decision-support platform.

7. REFERENCES

1. Russell, S. and Norvig, P. (2020), *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson.
2. Breiman, L. (2001), "Random Forests," *Machine Learning*, Vol. 45, pp. 5–32. DOI: 10.1023/A:1010933404324.
3. Pang, B. and Lee, L. (2008), "Opinion Mining and Sentiment Analysis," *Foundations and Trends in Information Retrieval*, Vol. 2, No. 1–2, pp. 1–135. DOI: 10.1561/1500000011.
4. Hochreiter, S. and Schmidhuber, J. (1997), "Long Short-Term Memory," *Neural Computation*, Vol. 9, No. 8, pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.
5. Murphy, K. P. (2022), *Probabilistic Machine Learning: An Introduction*, MIT Press.
6. Géron, A. (2022), *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd Edition, O'Reilly Media.
7. Achelis, S. B. (2000), *Technical Analysis from A to Z: Trading Strategies for the Investment Industry*, 2nd Edition, McGraw-Hill.
8. Graham, B. and Dodd, D. L. (2008), *Security Analysis: Principles and Technique*, 6th Edition, McGraw-Hill.
9. Hull, J. C. (2021), *Options, Futures, and Other Derivatives*, 11th Edition, Pearson.
10. Your project presentation, *AI-Powered Stock Screening and Analysis*, Academic Year 2026–2027 — used as the primary source for the proposed system architecture, modules, outputs, and project-specific implementation description.